

Reuben Chatterjee

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Experience

Applied Materials

Dec 2025 - Present

Data Scientist

- Architected and deployed a **multi model**, production grade **OCR pipeline** using **vision-language models** (Claude Sonnet 4.5, Qwen VL, PaddleOCR), achieving 85.7% precision across 500+ industrial part labels (printed, etched and engraved).
- Trained and evaluated an **RT-DETR** based **label detection system** by converting LabelMe annotations to COCO format, implementing custom IoU/containment matching, and achieving 97.8% precision on printed labels and 64.0% on engraved labels.
- Designed a comprehensive evaluation framework combining IoU with precision/recall/F1 metrics and visual debugging, while improving inference efficiency by 3-5x through **NVMe caching** and float16 GPU optimization on Databricks.

Alcamo Marketing

Jul 2025 - Oct 2025

Data Scientist

- **Reduced operational costs by 60% (\$20K annually)** by migrating data infrastructure from Adverity to Snowflake, building Snowflake tables and views with 100+ field mappings and transformations for 4 clients.
- Enabled **real-time reporting with 99% uptime** for 4 clients by **engineering 15 automated ETL pipelines** (Fivetran to Snowflake to Looker Studio) processing 50GB+ daily data from Google Ads, LinkedIn Ads, Meta, Salesforce, and HubSpot
- Engineered **ML-based customer segmentation** with clustering algorithms **achieving 90% accuracy scores**, analyzing **behavioral patterns across 100K+ customer records** to optimize engagement strategies and prioritize high-value opportunities

Ellis Lab, UC San Diego

Sep 2024 - Jun 2025

Graduate Research Assistant - Data Science

- Developed **automated data cleaning and validation pipeline** in R using Regex and NLP to standardize 5,000+ free-text responses, **reducing manual processing time by 90%**
- Applied **ANOVA, Tukey HSD** tests and correlation analysis to examine gender based differences male vs female dominant groups across team contributions, leadership roles, and project outcomes

Datamatics Global Services

Jun 2024 - Sep 2024

Data Scientist Intern

- Built **Python and SQL based ETL pipelines** to process 50GB+ of economic zone data for RAKEZ, enabling demand forecasting of industrial land lease activity
- Trained a Random Forest model to optimize zone planning decisions, achieving 92% accuracy and validating results via A/B testing

Projects

Afterpay Customer Retention Prediction Model

Python, XGBoost, LightGBM, Scikit-learn, Pandas, SHAP, Snowflake, SQL

- Built **gradient boosting churn prediction model** achieving **89% accuracy and 0.92 ROC-AUC** by engineering **50+ behavioral features** including RFM metrics, payment patterns, and temporal trends from 350K+ transaction records.
- Identified **\$2.1M in at-risk revenue** by **scoring 12,500 high-risk customers** through 4-tier risk segmentation system, enabling targeted retention campaigns that **improved customer lifetime value by 18%**.

Credit Card Anomaly Detection using Gradient Boosting

Python, SciKit-learn, Hugging Face, CNN, XGBoost, LightGBM, Random Forest, Seaborn

- Engineered 3,200+ behavioral features from 97,852 credit card transactions using domain-specific encodings and behavioral signals.
- Tuned LightGBM via multi-model comparison (RF, XGBoost, CatBoost), achieving 92% accuracy and 0.59 OOT AUC
- Reduced false positives by 10% via threshold tuning and SMOTE, contributing to \$2M+ projected annual savings

Education

University of California, San Diego

Master of Science in Data Science (CGPA:3.82)

Sep 2023 - Jun 2025

Courses: Statistical Models, Scalable Data Systems, Fraud & Pricing Analytics, Deep Learning

University of Mumbai

Bachelor of Engineering in Computer Engineering (CGPA:3.93)

Aug 2019 - May 2023

Courses: Data Structures, Big Data Analytics, DBMS, AI & Machine Learning, NLP

Technical Skills

Programming: Python, SQL, R, C++, JavaScript

Machine Learning & AI: PyTorch, Hugging Face Transformers, Langchain & LangGraph, LLM, VLM, OCR, Object Detection (RT-DETR), Model Evaluation (Precision/Recall/F1, IoU)

Data Engineering & MLOps: ETL & Batch Pipelines, Model Deployment & Inference (Databricks, AWS, Snowflake), GPU Optimization (float16, NVMe caching), Annotation Formatting (COCO, LabelMe), CLI, Git

Data Platforms & Tools: Snowflake, Databricks, JIRA, AWS, Azure, Hadoop, dbt, Docker, Fivetran

Visualization & Communication: Tableau, Looker Studio, Power BI, Excel, ggplot2, Shiny, Seaborn